The Fine-Structure Constant as an OPH Pixel Fixed Point

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Abstract

Conditional on a declared same-cell identification, Observer Patch Holography (OPH) assigns one screen cell two readings: an outside dimensionless area P and the electromagnetic observation scale seen from within. Their self-read equation is

$$P = \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P)},$$

where $A_{\mathrm{Th}}(P)$ is the Thomson-limit inverse coupling returned by a declared trial map. The contraction theorem gives existence and uniqueness, and outward-rounded interval certificates discharge its concrete hypotheses. The source map has the interval-certified unique root $\alpha^{-1} = 136.994835177413\ldots$. Adding the finite-screen gauge width inside the same loop gives the certified root $137.035660136946577\ldots$. The recommended experimental value is used only to test these source calculations. A physical low-energy prediction also requires the same-scheme hadronic spectral transport, target-independent selection of the applicable map, and a typed bridge proving that the two fixed-point coordinates read the same physical quantity. This construction supplies none of those premises.

Keywords: fine-structure constant, Observer-Patch Holography, pixel fixed point, electromagnetic coupling, Standard Model constants, hadronic vacuum polarization

What This Paper Contributes

The fine-structure constant is usually treated as an empirical constant. This paper studies it as a conditional branch readout of one OPH screen cell. The known physics is the low-energy Maxwell coupling, charged-lepton running, and low-energy hadronic spectral transport. OPH adds a declared pixel self-reference: conditional on a typed bridge proving that both expressions read one cell, the outside area coordinate P must agree with the inside electromagnetic observation scale.

The pixel map carries zero continuous dials. The fixed-point existence and uniqueness theorem used by the calculation, together with interval certificates, gives each declared map a unique fixed point: the source map contracts to $136.994835177413\ldots$ (3.0×10^{-4} relative from the measured endpoint) and the self-consistent gauge-width map to 137.035660136947 (2.5×10^{-6} relative; the declared map is certified, and the physical transport step is a separate premise). The residual gap has an explicit address: the hadronic transport needed in the same electromagnetic convention as the source map. This split is the contribution: a fixed-point equation for the pixel branch with no adjustable form, a visible address for the missing low-energy QCD step, and a clear boundary between a source theorem, the open map-selection and same-quantity premises, empirical endpoint input, and measured comparison.

1 Introduction

The screen cell in this calculation is not free-floating. On the declared carrier branch of the OPH framework it is a cell of the federated screen built from twelve-port icosahedral carriers, the same lineage whose port algebra witnesses the Standard Model gauge structure in the compact paper [2]; the local pixel equation reads back the electromagnetic resolution of one such cell.

Two declared principles carry the construction. The detuning law places the pixel ratio off the golden-ratio balance point by the Gaussian-normalized width of one electromagnetic observation, $P = \varphi + \sqrt{\pi}\,\alpha$. The constant-identification law is a declared closure premise: a typed bridge must prove that the fine-structure constant read inside the trial world and the substrate pixel readout are one quantity. Conditional on that bridge, the two principles leave one local variable and one self-read equation. Neither the exact fixed-point mathematics nor the source certificates construct the bridge or select between the two certified maps. The measured fine-structure constant enters only after the source calculation, as an endpoint test.

There is a long history of attempts to understand α from deeper structure. Jentschura and Nandori give a useful survey of first-principles attempts, including beta-function, symmetry, cutoff, and string-inspired ideas [6]. Golden-ratio constructions appear in historical geometric proposals [7] and in semi-empirical work on electroweak and flavor mixing [9]. Octonionic and exceptional-algebra approaches also produce values near 1/137 [8]. ArXiv papers use symbolic regression and quantum-information criteria to look for structure in Standard Model constants and electroweak parameters [10, 11]. The OPH calculation belongs to this broad search for structure. Its physical interpretation requires a target-independent choice of the map, a typed bridge proving that its input and output read the same local screen cell, and same-scheme transport to the laboratory endpoint.

Certified source roots. The executable source map contracts to the undressed source/root inverse coupling $\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots$, the interval-certified unique fixed point of the declared map (public interval certificate [4]; enclosure width 7.2×10^{-24}, $L \leq 0.0724$). The self-consistent map that adds the finite-screen unified gauge width $\alpha_U(P)$ inside the loop has the certified fixed point $137.035660136946577\ldots$, with relative residual 2.5×10^{-6}, about 1.6×10^4 measurement sigma. The combination $A^{\mathrm{fp}}_{\alpha_U} = 137.035959513609\ldots$ mixes the certified source root with $\alpha_U(P_{\mathrm{C}})$ evaluated at the CODATA-derived comparison pixel; it is not a fixed point of a single declared map. The measured value is a downstream endpoint test and does not fix either source root.

CODATA/NIST measured endpoint. CODATA/NIST reports the value 137.035999177(21) for $\alpha^{-1}(0)$. The difference from $A^{\mathrm{fp}}_{\alpha_U}$ is CODATA-minus-diagnostic bookkeeping; it is not the output of a same-scheme hadronic spectral calculation.

Intuitive picture: two source maps and one endpoint test. The three fine-structure numbers have different provenance. The first two are fixed points of declared source maps, while the third is the measured endpoint:

$$\text{source pixel} \longrightarrow \text{gauge-width dressed pixel} \longrightarrow \text{measured low-energy value},$$

or numerically,

$$136.994835177413\ldots \longrightarrow 137.035660136946577\ldots \longrightarrow 137.035999177(21).$$

The first two values are the interval-certified fixed points of the declared source map and of the declared self-consistent gauge-width map (public interval contraction certificate [4]). The root value $\alpha_{\text{root}}^{-1}$ is the inverse coupling returned by the incomplete source map before the low-energy charged vacuum is attached. The declared outside coordinate is the detuning $(P-\varphi)/\sqrt{\pi}$, and the declared inside coordinate is the electromagnetic observation strength at the same trial input. A typed same-quantity bridge and target-independent map selection are required before their fixed point can be interpreted as one cell quantity. Same-scheme endpoint transport is additionally required before that quantity can be compared as the laboratory fine-structure constant.

The contribution $\alpha_U(P_{\mathrm{C}})$ is the finite-screen unified gauge-width evaluated at the CODATA-derived comparison pixel. It is the main high-scale gauge-sector width carried by that pixel: the weak and color representation content must fit the pixel budget $P/4$, and the electroweak heat-kernel closure supplies the executable certificate. Adding this comparison-pixel width to the source root gives the mixed-provenance no-hadron diagnostic $A_{\alpha_U}^{\mathrm{fp}}$; that additive combination is a display packet, not a fixed point of any single declared map. The self-consistent reading, with $\alpha_U(P)$ evaluated inside the loop, is the certified fixed point $137.035660136946577\ldots$ quoted above. Both are close to the laboratory endpoint because the gauge width supplies almost all of the inverse-alpha gap from the naked source value.

The factor $C_{24,Q}$ is a back-solved calibration: the small endpoint dressing required after the gauge-width contribution is added, solved from the measured endpoint rather than computed from source data. It is near one because $\alpha_U(P_{\mathrm{C}})$ supplies the large part of the correction. Its excess $\alpha_U(P_{\mathrm{C}})(C_{24,Q}-1)$ is a comparison residual with the numerical size required of the residual same-scheme low-energy hadronic endpoint transport. It does not compute that transport. A zero-momentum laboratory photon does not probe an empty charged vacuum; it sees the Ward-projected $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and finite endpoint matching. In short, the observer measures the fully dressed low-energy coupling, while $A_{\alpha_U}^{\mathrm{fp}}$ is a mixed-provenance no-hadron diagnostic before that dressing.

The OPH background used here is spread across three papers. The observer-overlap starting point is the synthesis paper [1]. The source map and fixed-point branch are recorded in the compact overlap-consistency paper [2]. The charged-spectrum and particle-structure continuation is recorded in the particle paper [3]. The references give direct GitHub links to those source files.

Why this split matters: the fixed-point equation and the hadronic transport calculation answer different questions. The fixed-point equation tests closure for a declared local map. The hadronic calculation transports the electromagnetic current through low-energy QCD. A physical endpoint calculation must select the map without consulting the target, construct the same-cell bridge, and evaluate the same-scheme hadronic spectral functional from OPH source data.

Proposition 1 (Maxwell normalization of the endpoint branch)**.** *On the ordinary electromagnetic branch used by the fixed-point equation, the Ward-projected* $\mathrm{U}(1)_Q$ *channel carries*

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2(q^2;P){*}J_Q,$$

with

$$g_Q^2(q^2;P) = 4\pi\alpha_{\rm em}(q^2;P), \qquad A_{\rm Th}(P) = \alpha_{\rm em}^{-1}(0;P).$$

Thus the inside reading in the pixel equation is the Thomson-limit Maxwell coupling of the same electromagnetic current.

Proof. The compact OPH reconstruction identifies the unbroken electromagnetic branch as $\mathrm{U}(1)_Q$. On that abelian factor, the compact-gauge curvature reduces to $F_Q = dA_Q$, and the quadratic field action gives $d{*}F_Q = g_Q^2 {*}J_Q$. The particle-paper transport theorem then identifies the same $g_Q(q^2;P)$ with the Ward-projected running electromagnetic coupling. Taking $q^2 \to 0$ gives the endpoint quantity $A_{\rm Th}(P)$ used in the pixel fixed-point equation. □

The hadronic step uses the standard dispersion logic behind hadronic vacuum-polarization work. The relevant comparison data are electromagnetic spectral functions measured through $e^+e^- \to$ hadrons, as used in data-driven running-α and $g-2$ analyses [12, 13, 16]. In this paper, the decimal correction shown above is reported as the difference between the CODATA/NIST comparison value and the OPH pure source calculation. The source data in this paper emit no direct production calculation of the hadronic spectral function; a pure source endpoint theorem consumes that calculation as a premise.

2 Symbols

Symbol	Definition	Informal meaning
$\mathcal{A}(P)$	Patch algebra assigned to a screen patch P	The local observables available to one finite observer patch.
ω_P	State on $\mathcal{A}(P)$	The local physical description carried by that patch.
$\mathcal{A}(P \cap Q)$	Shared overlap algebra	The observables two neighboring patches can compare.
P	$a_{\text{cell}}/\ell_\star^2$	The dimensionless area of one screen cell in scale-readout units.
a_{cell}	Physical screen-cell area	The area of the local pixel on the holographic screen.
$\ell_\star$	OPH scale-certificate length	The length emitted by the selected no-G scale certificate.
ℓ_P	Planck length	The SI display of $\ell_\star$ after $G_{\text{SI}} = c^3\ell_\star^2/\hbar$.
φ	$(1+\sqrt{5})/2$	The self-similar entropy-balance point.
$\sqrt{\pi}$	Boundary Gaussian normalization width	The conversion factor from pixel displacement to observation width.
$\alpha_{\text{ext}}(P)$	$(P-\varphi)/\sqrt{\pi}$	The declared outer detuning coordinate.
$A_{\text{Th}}(P)$	$\alpha_{\text{em}}^{-1}(0;P)$	The inverse electromagnetic endpoint returned by the declared trial map.
$\alpha_{\text{in}}(P)$	$1/A_{\text{Th}}(P)$	The declared inner coordinate; its same-cell interpretation requires a typed bridge.
E_P	Planck energy	The energy unit associated with the selected scale certificate. The numerical equations use Planck units unless GeV labels are attached.
$M_U(P)$	$E_P \mathrm{e}^{-2\pi} P^{1/6}$	The OPH unification-scale readout returned at the declared trial coordinate.
$E_{\text{cell}}(P)$	$E_P/\sqrt{P}$	The local cell energy readout.
$\alpha_U(P)$	Unified coupling solved from the heat-kernel closure equation	The coupling at $M_U(P)$ that makes the gauge representation entropy match the pixel.
$\alpha_i(\mu;P)$	Gauge coupling $i = 1,2,3$ at scale μ	The hypercharge, weak, and color couplings emitted by the same trial P.

Symbol	Definition	Informal meaning
b_i	$(33/5, 1, -3)$	The one-loop MSSM beta coefficients used as the high-scale running convention here; a declared structural selection.
$m_Z(P)$	Self-consistent electroweak anchor scale	The Z-scale generated by $v(P)$, $\alpha_1(P)$, and $\alpha_2(P)$.
$v(P)$	$E_{\mathrm{cell}}(P)\exp[-2\pi/(\beta_{\mathrm{EW}}\alpha_U(P))]$	The electroweak transmutation scale.
β_{EW}	$N_c + 1 = 4$ for $N_c = 3$	The coefficient that controls the exponential drop to the electroweak scale.
$A_Z(P)$	$\alpha_{\mathrm{em}}^{-1}(m_Z^2; P)$	The electroweak-scale electromagnetic anchor.
$\Delta_{\mathrm{lep}}(P)$	Lepton transport contribution to inverse alpha	The exact one-loop kernel evaluated on the frozen phenomenological e, μ, τ spectrum.
$\Delta_{\mathrm{had}}(P)$	Hadronic transport contribution	The low-energy QCD spectral contribution needed for the pure source calculation.
$\Delta_{\mathrm{EW}}(P)$	Electroweak/scheme endpoint remainder	Same-scheme finite remainder needed to connect the source anchor to the endpoint.
$\Delta_{\mathrm{calc}}(P)$	Calculated charged-fermion transport contribution	The exact kernel on the supplied lepton continuation plus the screened quark continuation; not a pure source prediction.
$\Delta^{\mathrm{fp}}_{\mathrm{H,cal}}$	Root-to-CODATA bookkeeping difference	The inverse-alpha amount between the naked root value and the CODATA/NIST central endpoint; it includes the comparison-pixel $\alpha_U(P_{\mathrm{C}})$ contribution and is not the post-α_U hadronic gap.
$\Delta^{\mathrm{fp}}_{\mathrm{H,req}}$	Required hadronic endpoint correction after $A^{\mathrm{fp}}_{\alpha_U}$	The inverse-alpha amount by which the displayed OPH output misses the CODATA/NIST central endpoint.
$R_Q(P)$	$A_{\mathrm{Th}}(P) - [A_Z(P) + \Delta_{\mathrm{calc}}(P)]$	The residual same-scheme hadronic endpoint contribution.
$A^{\mathrm{fp}}_{\alpha_U}$	$\alpha_{\mathrm{root}}^{-1} + \alpha_U(P_{\mathrm{C}})$	Mixed-provenance no-hadron comparison diagnostic; the root is source-side and P_{C} is CODATA-derived.

Symbol	Definition	Informal meaning
$J_{24,Q}(P)$, $\omega_Q(P)$, $\Xi_Q(P)$	Source-side hadronic spectral objects required for a final endpoint theorem	The Jacobi/Stieltjes spectral payload, normalization, and finite same-scheme remainder that must be emitted without looking at the Thomson target.
P_{C}	CODATA/NIST comparison pixel	The pixel obtained from the CODATA/NIST Thomson endpoint under the OPH outer equation.

3 Shared Collar Convention for χ_ν

The χ_ν protected-reserve theorem uses the public endpoint branch convention

$$P_\chi = P_{\mathrm{C}} = 1.630968209403959\ldots.$$

Its scalar reserve density is

$$\epsilon_{\mathrm{res}} = \frac{P_\chi}{24}$$

only after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass in the screen microphysics and susceptibility papers. This does not make $P/4$ a primitive Hilbert-space dimension. In this paper, $P/4$ is the local screen-cell entropy budget used by the electromagnetic fixed-point branch; $P_\chi/24$ is the shared scalar-reserve density used by the χ_ν collar branch.

4 Step 1: The OPH Starting Rule

OPH starts from a finite screen covered by observer patches. If P_1 and P_2 overlap, then their induced states must agree on the shared algebra:

$$\omega_{P_1}|_{\mathcal{A}(P_1\cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1\cap P_2)}. \tag{1}$$

Informally: no observer sees the whole world. Physics is what survives when neighboring local descriptions can be checked against each other and made consistent on the parts they share.

The quantitative fine-structure branch applies this rule to a single local screen cell. The cell has one outside description and one inside description. The outside description is geometric. The inside description is electromagnetic. A declared typed bridge identifies both as readings of one cell; the fixed-point theorem is conditional on that premise.

Why this step is needed: OPH does not start by assigning constants to nature. It starts by asking what finite observers can compare. A dimensionless constant can be derived only if it is the fixed value of such a comparison.

5 Step 2: The Pixel Variable

Define the local pixel ratio

$$P := \frac{a_{\text{cell}}}{\ell_\star^2}. \tag{2}$$

Here a_{cell} is the screen-cell area and $\ell_\star^2$ is the area scale emitted by the separate OPH gravity readback fixed point.

Informally: P says how large the screen cell is when measured in scale-readout units. It is dimensionless, so it can be compared directly with pure numbers such as φ and $\sqrt{\pi}$. After the gravity row emits G, the same scale is displayed as the Planck area in SI units. The pixel equation fixes this dimensionless ratio; it does not determine the SI scale by itself.

Why this step is needed: α has no units. A unitless electromagnetic number has to be matched to a unitless geometric number. Dividing the cell area by the emitted scale area gives that number without using the pixel equation to derive G.

6 Step 3: The Golden-Ratio Balance

The self-similar balance point is

$$\varphi = 1 + \frac{1}{\varphi}, \qquad \varphi^2 - \varphi - 1 = 0, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{3}$$

Informally: φ is the fixed point of the simplest self-similar split, where the whole-to-large ratio equals the large-to-small ratio. OPH takes it as the declared zero-detuning balance point for the local pixel.

The pixel does not sit exactly at φ. It sits at

$$\Delta_P := P - \varphi. \tag{4}$$

Informally: Δ_P is the small amount by which the realized cell is displaced from exact self-similar equilibrium. That small displacement is what becomes the electromagnetic observation strength.

Why this step is needed: a fixed point needs a reference position. The detuning principle declares the golden ratio as the self-similar reference, while $P - \varphi$ measures how far the actual cell sits away from it.

7 Step 4: Boundary Gaussian Normalization

The boundary normalization converts Δ_P into the outside coupling readout:

$$\alpha_{\text{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}. \tag{5}$$

Informally: the boundary Gaussian width normalizes the screen displacement. Dividing by $\sqrt{\pi}$ turns the geometric displacement into a dimensionless observation strength. The Gaussian choice belongs to the declared detuning principle.

Why this step is needed: the outside screen variable is an area displacement. The inside electromagnetic variable is a coupling. The declared boundary normalization supplies the proposed conversion between those two readings.

8 Step 5: The Inside Electromagnetic Readout

Let

$$A_{\mathrm{Th}}(P) := \alpha_{\mathrm{em}}^{-1}(0; P) \tag{6}$$

be the inverse electromagnetic coupling at zero momentum returned by the declared map at trial pixel P. The inside coupling is

$$\alpha_{\mathrm{in}}(P) := \frac{1}{A_{\mathrm{Th}}(P)}. \tag{7}$$

Informally: an observer inside the encoded world does not see P as a screen area. The observer sees the strength of electromagnetism. Since particle physicists quote the low-energy coupling as an inverse number near 137, the actual coupling is $1/A_{\mathrm{Th}}(P)$.

Why this step is needed: the measured fine-structure constant is the Thomson-limit coupling. The calculation must therefore end at zero momentum, after the electroweak anchor has been transported through all charged degrees of freedom.

9 Step 6: The Fixed-Point Equation

Conditional on the same-invariant bridge, closure requires the outside and inside couplings to agree:

$$\alpha_{\mathrm{ext}}(P) = \alpha_{\mathrm{in}}(P). \tag{8}$$

Using Eqs. (5) and (7),

$$\frac{P - \varphi}{\sqrt{\pi}} = \frac{1}{A_{\mathrm{Th}}(P)}. \tag{9}$$

Equivalently,

$$\boxed{H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P)} = 0} \tag{10}$$

or

$$\boxed{P = G(P) := \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P)}.} \tag{11}$$

Informally: feed a trial pixel into the declared source chain. It returns a Thomson endpoint. If the bridge proves that endpoint and the detuning are two readings of one cell, closure selects a value that comes back unchanged.

Why this step is needed: a trial value is not enough. Once the bridge proves that both readings belong to one cell, self-consistency requires them to agree. That is why the conditional answer is a fixed point rather than a one-way evaluation.

10 Step 7: The Source Map

For a trial P, the source map first emits

$$M_U(P) = E_P\, \mathrm{e}^{-2\pi}\, P^{1/6}, \tag{12}$$

$$E_{\text{cell}}(P) = \frac{E_P}{\sqrt{P}}. \tag{13}$$

Informally: $M_U(P)$ is the high-scale source readout, and $E_{\text{cell}}(P)$ is the declared local-energy coordinate. Both are functions of the same trial input P. Interpreting that input as one physical screen cell requires the same-cell bridge.

Why this step is needed: the inside electromagnetic coupling has to be generated from the same P used in the outside equation. These two formulas begin that inside calculation without adding a separate high-scale fit parameter.

11 Step 8: Electroweak Transmutation

The source branch uses

$$\beta_{\text{EW}} = N_c + 1 = 4 \tag{14}$$

with $N_c = 3$. The same value has a representation-theoretic reading in the Standard Model gauge paper: the selected exterior matter package contains exactly four weak doublets, $3 + 1$ from the quark and lepton sectors of one generation, so $N_c + 1$ and the weak-doublet multiplicity name one coefficient. The identification of that abstract load with the physical screen load is the separate common-carrier receipt. The branch defines

$$v(P, \alpha_U) = E_{\text{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\text{EW}}\alpha_U}\right]. \tag{15}$$

Informally: the weak scale is exponentially lower than the cell scale. The unified coupling controls that descent, so changing α_U changes the electroweak scale sharply.

Why this step is needed: the fine-structure constant is measured at low energy, and the source map begins at the cell scale. The transmutation formula explains how the electroweak scale is produced from the same source data. After independent direct public-record closure, a positive refinement-natural identification of the screen and electroweak load carriers gives the conditional bridge $N_{\text{bridge}} = \pi \exp[6\pi/(P\alpha_U)]$, certified by interval arithmetic at 3.53×10^{122}. The bridge tests the capacity; it does not construct the public-record readback.

12 Step 9: One-Loop Gauge Running

For $i = 1, 2, 3$, define

$$(b_1, b_2, b_3) = \left(\frac{33}{5},\, 1,\, -3\right), \tag{16}$$

and run the couplings by

$$\alpha_i^{-1}(\mu; P, \alpha_U) = \alpha_U^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right). \tag{17}$$

The triple $(33/5, 1, -3)$ is the one-loop MSSM beta-coefficient set. Its use is a declared structural selection: the Standard Model coefficient triple has no pixel-residual root in the declared scan, no supersymmetric threshold matching enters the map, and the convention is a counted structural selection.

Informally: once α_U is guessed, all three gauge couplings at lower scales are fixed. The same high-scale coupling determines α_1, α_2, and α_3.

Why this step is needed: the electromagnetic coupling is a mixture of the weak and hypercharge couplings. Running the gauge couplings supplies the ingredients needed to form that mixture at the Z-scale.

13 Step 10: The Self-Consistent Z-Scale

The hypercharge coupling in Standard Model normalization is

$$\alpha_Y(\mu; P) = \frac{3}{5}\alpha_1(\mu; P). \tag{18}$$

The tree-level Z-mass readout is

$$m_Z(\mu; P, \alpha_U) = \frac{v(P, \alpha_U)}{2}\sqrt{4\pi\alpha_2(\mu; P, \alpha_U) + 4\pi\alpha_Y(\mu; P, \alpha_U)}. \tag{19}$$

The source point uses the self-consistency condition

$$\mu = m_Z(\mu; P, \alpha_U). \tag{20}$$

Informally: the Z-scale is determined inside the calculation. It is the scale at which the running couplings and the electroweak transmutation formula reproduce their own Z-mass readout.

Why this step is needed: using a measured m_Z here would hide experimental input inside the source map. The self-consistency condition keeps the source calculation closed.

14 Step 11: Heat-Kernel Gauge Closure

For SU(2), the irreducible representations are labeled by $j = n/2$, $n = 0, 1, \ldots, N_2$, with

$$d_j = 2j + 1, \qquad C_2(j) = j(j+1). \tag{21}$$

For SU(3), representations are labeled by highest weights (p, q), $0 \le p, q \le N_3$, with

$$d_{p,q} = \frac{(p+1)(q+1)(p+q+2)}{2}, \tag{22}$$

$$C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}. \tag{23}$$

For a compact group G in this finite representation cutoff, define

$$Z_G(t) = \sum_R d_R \, \mathrm{e}^{-tC_2(R)}, \tag{24}$$

$$\bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R \, \mathrm{e}^{-tC_2(R)} \log d_R. \tag{25}$$

The heat-kernel source parameters are

$$t_2 = 4\pi^2\alpha_2(m_Z; P, \alpha_U), \qquad t_3 = 4\pi^2\alpha_3(m_Z; P, \alpha_U). \tag{26}$$

The pixel-closure equation is

$$\boxed{\bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3) = \frac{P}{4}.} \tag{27}$$

For a trial P, Eq. (27) is solved for $\alpha_U(P)$. The closure sums the SU(2) and SU(3) representation towers only; the selection that hypercharge carries no representation entropy against the $P/4$ budget is a declared choice. The repair round count used by the finite current construction counts the $\mathfrak{u}(1)$ generator $(8+3+1=12)$; the two conventions are distinct declared choices.

Informally: the representation entropy carried by the weak and color sectors must match the pixel capacity assigned to the cell. This step is what locks the unified coupling to the same P that appears in the outer equation.

Why this step is needed: α_U should not be chosen by hand. The heat-kernel closure turns the finite representation content of the gauge sectors into an equation for $\alpha_U(P)$.

15 Step 12: The Electroweak Anchor

Once $\alpha_U(P)$ and $m_Z(P)$ are solved, define

$$\alpha_Y(m_Z; P) = \frac{3}{5}\alpha_1(m_Z; P), \tag{28}$$

$$\alpha_{\mathrm{em}}(m_Z^2; P) = \left(\frac{1}{\alpha_2(m_Z; P)} + \frac{1}{\alpha_Y(m_Z; P)}\right)^{-1}. \tag{29}$$

The source-locked electroweak anchor is

$$\boxed{A_Z(P) := \alpha_{\mathrm{em}}^{-1}(m_Z^2; P).} \tag{30}$$

The weak mixing readout is

$$\sin^2\theta_W(m_Z; P) = \frac{\alpha_{\mathrm{em}}(m_Z^2; P)}{\alpha_2(m_Z; P)}. \tag{31}$$

Informally: the source map has reached the electromagnetic coupling at the electroweak anchor scale. The low-energy 1/137 number appears only after this anchor is transported to zero momentum.

Why this step is needed: $A_Z(P)$ is the clean place where the source map meets standard electroweak physics. From here the problem becomes a transport problem for the electromagnetic current.

16 Step 13: Frozen Charged-Spectrum Continuation Used by the Transport

The one-loop kernel below is exact for supplied fermion masses, but the charged spectrum supplied to it in this section is a phenomenological continuation, not an OPH-derived mass theorem. It uses $N_c = 3$, $N_g = 3$, and the additional assumptions

$$\epsilon = \frac{1}{6}, \qquad \delta = \frac{\beta_{\mathrm{EW}}}{2N_c N_g} = \frac{2}{9}. \tag{32}$$

No physical charged-spectrum derivation is claimed here; such a derivation requires a source-to-mass attachment, a phase-transport law fixing $\delta = 2/9$, and an attachment of the ordered roots to physical charged-family lines. For a positive-semidefinite square-root-mass carrier

$$C = aI + \rho(e^{i\delta}R + e^{-i\delta}R^2), \qquad Q = \frac{1 + 2(\rho/a)^2}{3},$$

so $\rho/a = 1/\sqrt{2}$ is exactly equivalent to the empirical Koide value $Q = 2/3$. At balance this identity is physical throughout the positive chamber $|\delta| \le \pi/12 \pmod{2\pi/3}$, and it does not select the phase.

The finite balance normalization has a closed GNS construction. On $\mathcal{V}_K = \mathbf{1} \oplus \chi \oplus \bar{\chi}$ with a minimal two-state orientation record, take the connected algebra $B(\mathcal{V}_K \otimes \mathbb{C}^2) \simeq M_6(\mathbb{C})$ and event $E_+ = P_0 \otimes I_2 + P_c \otimes q_+$. Its singlet and oriented charged blocks both have rank two. Born–Lüders conditioning of $I_6/6$ gives $p_0 = p_c = 1/2$, or $S_c - S_0 = \ln 2$. In the tracial GNS space, $e_0, e_+, e_- \mapsto I, R, R^2$ is unitary, and the canonical square-root amplitude gives

$$a = \sqrt{p_0}, \qquad \rho = \sqrt{p_c/2}, \qquad \frac{\rho}{a} = \frac{1}{\sqrt{2}}.$$

The connected register fixes the normalization left free by the direct-sum MaxEnt model. Minimality of this response-local register is an explicit declared hypothesis. A physical charged-lepton conclusion requires a chiral regular-C_3 carrier and faithful trace-preserving unital completely positive maps Φ, Ψ between the source and physical response algebras, with $\Psi\Phi = \mathrm{id}$. If they intertwine the block records and identify the physical GNS vector with $C = (Y_e^\dagger Y_e)^{1/4}$, Kadison–Schwarz makes Φ a $*$-monomorphism and an exact L^2 isometry. It therefore carries the equal source block powers to the physical response. For an accepted finite chiral checkpoint with three left and three right family modes, positive kinetic metrics, and a committed neutral Higgs direction, $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$. With $M_F = X_F^2$, this gives

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

The extended branch $\mathrm{OPH}^+_{\mathrm{ch}}$ adds graded physical completion and quotient source-law selection. Given an exhaustive declared carrier class with a positive winner gap, a source-closed BV/BRST continuum, and an interval or contraction certificate for the charged QFT self-map, it selects one dressed mass readout. A balanced C_3 fixed point gives $Q = 2/3$; a C_3-symmetric fixed point with charged attenuation $\chi_\star$ gives

$$Q = \frac{1 + e^{-2\chi_\star}}{3}.$$

An off-plane response requires the full response operator. These are additional branch conditions beyond the three axioms. No 12/24 receipt supplies that physical attachment. Define three Koide roots

$$r_k = 1 + \sqrt{2}\cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2, \tag{33}$$

then sort them in increasing order and write the sorted list as (r_1, r_2, r_3).

The quark exponent vectors are

$$\boldsymbol{n}_u = (2N_c, N_c, 0) = (6, 3, 0), \qquad \boldsymbol{n}_d = (2N_c, N_c + 1, N_c - 1) = (6, 4, 2). \tag{34}$$

With $v = v(P)$,

$$m_u = \frac{v}{\sqrt{2}}\epsilon^6, \qquad m_c = \frac{v}{\sqrt{2}}\epsilon^3, \qquad m_t = \frac{v}{\sqrt{2}}, \tag{35}$$

$$m_d = \frac{v}{\sqrt{2}}\epsilon^6, \qquad m_s = \frac{v}{\sqrt{2}}\epsilon^4, \qquad m_b = \frac{v}{\sqrt{2}}\epsilon^2. \tag{36}$$

For charged leptons the exponent vector is

$$\boldsymbol{n}_e = (7, 4, 3). \tag{37}$$

This vector and the normalization below are declared phenomenological transport inputs. The OPH axioms do not derive $\boldsymbol{n}_e = (7, 4, 3)$ or the resulting determinant identity $\det M_e = v(P)^3/(2\,6^{14})$. Every common shift $(7+k, 4+k, 3+k)$ has the same ratio residual while changing the determinant. The extra factor $2^{1/6}$ below is asserted rather than derived. Define

$$\log g_c = \frac{1}{3}\sum_{a=1}^{3} \log\left(\frac{r_a^2\sqrt{2}\,6^{n_{e,a}}}{v}\right), \qquad s_0 = \mathrm{e}^{-\log g_c}, \qquad s_e = s_0\, 2^{1/6}. \tag{38}$$

Then

$$m_e = s_e r_1^2, \qquad m_\mu = s_e r_2^2, \qquad m_\tau = s_e r_3^2. \tag{39}$$

Equations (38)–(39) define a transport input, not a source-derived charged-lepton mass law. No charged RG/threshold theorem maps this coordinate to physical pole masses.

Informally: this section specifies a phenomenological spectrum used by the transport diagnostic. The lepton vacuum-polarization kernel is an exact one-loop calculation conditional on those masses; that exactness does not determine the masses themselves. The quark part is also a perturbative continuation. The confined hadronic QCD spectral measure is a separate low-energy object.

Why this step is needed: vacuum polarization depends on charged particles. The transport diagnostic cannot be evaluated until a charged spectrum and its charges are supplied. A pure OPH source theorem would first have to replace the continuation spectrum by independently derived charged masses.

17 Step 14: Exact One-Loop Fermion Transport

For a fermion of mass m_f, electric charge Q_f, and multiplicity N_f, define

$$K_f(Q^2; m_f, Q_f, N_f) = \frac{2N_f Q_f^2}{\pi}\int_0^1 x(1-x)\log\left(1 + \frac{Q^2 x(1-x)}{m_f^2}\right) \mathrm{d}x. \tag{40}$$

The integral has the following closed form. Let

$$z = \frac{Q^2}{m_f^2}, \qquad a = \frac{z}{4}. \tag{41}$$

Then

$$\int_0^1 x(1-x)\log(1 + zx(1-x))\,\mathrm{d}x = -\frac{5}{18} + \frac{1}{6a} + \frac{(2a-1)\sqrt{1+a}\,\operatorname{asinh}(\sqrt{a})}{6a^{3/2}}, \tag{42}$$

with

$$\operatorname{asinh}(\sqrt{a}) = \log(\sqrt{a} + \sqrt{1+a}). \tag{43}$$

The lepton contribution is

$$\Delta_{\text{lep}}(P) = K_e(m_Z(P)^2; m_e, 1, 1) + K_\mu(m_Z(P)^2; m_\mu, 1, 1) + K_\tau(m_Z(P)^2; m_\tau, 1, 1). \tag{44}$$

The naive five-quark contribution is

$$\begin{aligned}\Delta_q^{\text{naive}}(P) = {} & K_u(m_Z(P)^2; m_u, 2/3, 3) + K_d(m_Z(P)^2; m_d, -1/3, 3) \\ & + K_s(m_Z(P)^2; m_s, -1/3, 3) + K_c(m_Z(P)^2; m_c, 2/3, 3) + K_b(m_Z(P)^2; m_b, -1/3, 3).\end{aligned} \tag{45}$$

The declared hadronic screening ansatz is

$$S_{\text{calc}}(P) = 1 - \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{46}$$

The form of S_{calc} is a declared selection of the quark continuation and carries no perturbative derivation: the leading perturbative correction to quark vacuum polarization is the enhancement $R \to R_0(1 + \alpha_s/\pi + \cdots)$ with no N_c factor, about +3.7 percent at m_Z, while $S_{\text{calc}} \approx 0.887$ is an 11 percent suppression standing in for confinement cutting off the light-quark logarithms. Thus

$$\Delta_{\text{calc}}(P) = \Delta_{\text{lep}}(P) + S_{\text{calc}}(P)\Delta_q^{\text{naive}}(P). \tag{47}$$

Informally: the supplied charged-lepton continuation is propagated through an exact one-loop kernel. Quarks are treated by the one-loop kernel times the declared screening ansatz. The expected gap is the confined hadronic spectral transport plus the matching terms needed to use the same electromagnetic-current convention all the way to the endpoint.

Why this step is needed: the Thomson endpoint is not the same as the Z-scale anchor. Charged particles screen the electromagnetic current between those scales, and the kernel is the mathematical form of that screening.

18 Step 15: The Source Calculation Endpoint

The calculated endpoint is

$$A_{\text{calc}}(P) = A_Z(P) + \Delta_{\text{calc}}(P). \tag{48}$$

Putting A_{calc} into Eq. (11) gives the source map

$$G_{\text{calc}}(P) = \varphi + \frac{\sqrt{\pi}}{A_{\text{calc}}(P)}. \tag{49}$$

The numerical solve uses

$$G_{\text{calc}}(P_{\text{root}}) = P_{\text{root}} \tag{50}$$

and emits

$$P_{\text{root}} = 1.63097209585889737696451390350695562847\,9\ldots, \tag{51}$$

$$\alpha_{\text{root}}^{-1} = 136.99483517741293729528942946443690285 76\ldots. \tag{52}$$

The fixed point is interval-certified as existing and unique on its interval (interval certificate, available in the public artifact bundle [4]; enclosure widths 6.8×10^{-28} in P and 7.2×10^{-24} in α^{-1}, Lipschitz bound $L \leq 0.0724$). The source anchor at the certified root point is

$$A_Z(P_{\text{root}}) = 128.30826805798759734790405761408475887503\ldots. \tag{53}$$

The calculated transport contribution is

$$\Delta_{\text{calc}}(P_{\text{root}}) = 8.68656711942533994738537185035214398252\ldots. \tag{54}$$

Informally: the fixed-point algebra lands at a stable value near 137, with no measured alpha inserted into the solve. The residual gap has a precise address: low-energy hadronic transport in the same endpoint convention.

Why this step is needed: this is the non-circular check. It shows what the OPH source chain produces before the calibrated hadronic endpoint correction is added.

19 Step 16: The CODATA/NIST Measured Endpoint and Comparison Pixel

The CODATA/NIST 2022 inverse fine-structure constant is

$$A_{\text{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{55}$$

with concise form 137.035999177(21). The corresponding comparison pixel is

$$P_{\text{C}} = \varphi + \frac{\sqrt{\pi}}{A_{\text{C}}} = \begin{array}{l}\texttt{1.6309682094039593248792798477826489413359828516279250606661}\\ \texttt{507533907793398933432}\end{array} \tag{56}$$

The CODATA/NIST central coupling is

$$\alpha(0) = \frac{1}{A_{\text{C}}} = \begin{array}{l}\texttt{0.0072973525643314250302457952646916832280660213133653604957}\\ \texttt{7988038199335615736399 28}\end{array} \tag{57}$$

Informally: once the measured Thomson endpoint is supplied, the outer OPH equation fixes the corresponding comparison pixel immediately. It is the direct readout of one equation.

Why this step is needed: the measured endpoint and the comparison pixel are two algebraic forms of the same comparison definition. Reporting both makes the measured provenance of the pixel explicit; it does not identify that coordinate with a source-selected screen cell.

Two-P provenance audit. There are two distinct pixel numerals in this paper. The source-only solve gives P_{root} in Eq. (51); it is the branch that audits what the OPH source chain emits before a same-scheme hadronic endpoint correction is supplied. The comparison value P_{C} in Eq. (56) is different by about 3.9×10^{-6} in P-space because it is defined from the measured CODATA/NIST Thomson endpoint. Thus the published comparison pixel is a measured-endpoint display, not a derivation of α. Downstream quantities that use P_{C} inherit that measured endpoint by definition; downstream quantities that use P_{root} are source-branch diagnostics.

20 Step 17: Endpoint Accounting at the CODATA/NIST Comparison Pixel

At $P = P_\mathrm{C}$, the source point gives

$$A_Z(P_\mathrm{C}) = \begin{array}{l}\texttt{128.30796547328624820996110874175671618724547618036535646005}\\ \texttt{34216963511778416828564407872472 8}\end{array} \tag{58}$$

$$\Delta_\mathrm{calc}(P_\mathrm{C}) = \begin{array}{l}\texttt{8.68656784270852840098544254288596976826722173764873642 33784}\\ \texttt{57738999378445910678308096117959 0}\end{array} \tag{59}$$

The transport required by the CODATA/NIST measured endpoint is

$$\Delta_\mathrm{req}(P_\mathrm{C}) = A_\mathrm{C} - A_Z(P_\mathrm{C}) = \begin{array}{l}\texttt{8.72803370371375179003889125824328381275452381963464353 99465}\\ \texttt{78303648822158317143559212752 72}\end{array} \tag{60}$$

Therefore the source-side residual at the CODATA/NIST comparison pixel is

$$\begin{aligned} R_Q(P_\mathrm{C}) &= \Delta_\mathrm{req}(P_\mathrm{C}) - \Delta_\mathrm{calc}(P_\mathrm{C}) \\ &= \begin{array}{l}\texttt{0.04146586100522338905344871535731404448730208198590711 65681}\\ \texttt{20564649443712406465251116634 76}\end{array} \end{aligned} \tag{61}$$

Informally: the CODATA/NIST measured value differs from the pure source calculation by a small, precisely localized inverse-alpha contribution. The golden-ratio equation, the heat-kernel closure, and the numerical fixed-point solve all point to the same address for the difference: the same-scheme low-energy hadronic transport.

Why this step is needed: this accounting prevents the hadronic contribution from being hidden inside the final number. It shows the source anchor, the calculated transport, the required transport, and the residual in the same units.

21 Step 18: The OPH Output and the Required Hadronic Correction

The fixed-point calculation emits

$$A_\mathrm{calc}(P_\mathrm{root}) = \alpha_\mathrm{root}^{-1} = \begin{array}{l}\texttt{136.99483517741293729528942946443690285756120615103539318450}\\ \texttt{2}\end{array}. \tag{62}$$

The interval certificate and continuous-integration identity test are included in the public artifact bundle [4]. At the CODATA-derived comparison pixel, the finite-screen unified gauge-width contribution is

$$\alpha_U(P_\mathrm{C}) = 0.041124336195630495.$$

Combining that comparison-pixel gauge width with Equation (62) gives the mixed-provenance no-hadron diagnostic

$$A_{\alpha_U}^\mathrm{fp} = A_\mathrm{calc}(P_\mathrm{root}) + \alpha_U(P_\mathrm{C}) = 137.035959513608567790289429464437\ldots, \tag{63}$$

or

$$\alpha_{\mathrm{src}+U}^\mathrm{fp} = \left(A_{\alpha_U}^\mathrm{fp}\right)^{-1} = 0.0072973546764613521328727595 9657\ldots. \tag{64}$$

This $A^{\mathrm{fp}}_{\alpha_U}$ value is comparison bookkeeping, not a source-only prediction: $\alpha^{-1}_{\mathrm{root}}$ is evaluated on the source root, while $\alpha_U(P_{\mathrm{C}})$ inherits the measured endpoint through P_{C}. It is a mixed-provenance display packet (inner value from the certified source root plus α_U at the CODATA-derived comparison pixel); it is not a fixed point of any single declared map. The certified self-consistent gauge-width fixed point is $\alpha^{-1} = 137.035660136946577\ldots$.

Compared with the CODATA/NIST central measured endpoint,

$$A_{\mathrm{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{65}$$

the CODATA-minus-diagnostic bookkeeping gap is

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = A_{\mathrm{C}} - A^{\mathrm{fp}}_{\alpha_U} = \texttt{0.0000396633914322097105705355563} \quad \ldots. \tag{66}$$

Equivalently,

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = 0.0000396634 \pm 0.000000021, \qquad \frac{\Delta^{\mathrm{fp}}_{\mathrm{H,req}}}{A_{\mathrm{C}}} = 2.89438 \times 10^{-7}. \tag{67}$$

This is the inverse-alpha gap between the mixed diagnostic and the CODATA/NIST central value. Its physical address is the low-energy QCD/hadronic vacuum-polarization and same-scheme transport of the Ward-projected electromagnetic current, but the subtraction does not calculate that transport. The 2.89438×10^{-7} ratio describes the mixed diagnostic only. The distinct self-consistent gauge-width map has fixed point $\alpha^{-1} = 137.035660136946577\ldots$ and relative residual 2.5×10^{-6}, about 1.6×10^4 measurement sigma.

For bookkeeping, the full root-to-CODATA difference is

$$\begin{aligned}
\Delta^{\mathrm{fp}}_{\mathrm{H,cal}} &= A_{\mathrm{C}} - A_{\mathrm{calc}}(P_{\mathrm{root}}) \\
&= \texttt{0.04116399958706270471057053556309714243\,9} \\
&= \alpha_U(P_{\mathrm{C}}) + \Delta^{\mathrm{fp}}_{\mathrm{H,req}}.
\end{aligned} \tag{68}$$

This larger difference is bookkeeping: it contains the comparison-pixel unified gauge-width contribution and the CODATA-minus-diagnostic gap.

Equivalently, the CODATA/NIST central value can be written as

$$A_{\mathrm{C}} = \alpha^{-1}_{\mathrm{root}} + \alpha_U(P_{\mathrm{C}})\, C_{24,Q}, \qquad C_{24,Q} = 1.000964474933857388040135711404 5152\ldots. \tag{69}$$

The factor $C_{24,Q}$ is a back-solved calibration: compact comparison accounting, solved from the measured endpoint, for the finite same-scheme hadronic endpoint transport needed for source-only closure. Equivalently, $\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = \alpha_U(P_{\mathrm{C}})(C_{24,Q} - 1)$. The factor is not a source-emitted hadronic/QCD payload.

Here A_{C} is the CODATA/NIST 2022 inverse fine-structure constant [19, 20]. The source value $A_{\mathrm{calc}}(P_{\mathrm{root}}) = \alpha^{-1}_{\mathrm{root}}$ is a reproducible source-root output [4]; $A^{\mathrm{fp}}_{\alpha_U}$ mixes that root with $\alpha_U(P_{\mathrm{C}})$. Equation (66) is the measured-comparison residual in inverse-alpha units.

Informally: the mixed comparison diagnostic is $137.035959513609\ldots$; the certified self-consistent gauge-width fixed point is $137.035660136946577\ldots$ (2.5×10^{-6} relative); and the CODATA/NIST measured value is $137.035999177(21)$. The residual difference is the size required of the absent same-scheme endpoint contribution; its source derivation from a same-scheme hadronic spectral backend is a premise this paper does not supply.

This marks the calculation boundary: additional digits require a source-derived hadronic spectral calculation.

The endpoint decomposition is

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P). \tag{70}$$

The calculated transport contains the exact one-loop charged-lepton contribution and a structured quark-screening continuation. The residual pure source object can be written compactly as

$$R_Q(P) = \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P) - \left[\Delta_{\mathrm{calc}}(P) - \Delta_{\mathrm{lep}}(P)\right], \tag{71}$$

where the right side is understood in the same renormalization and endpoint scheme as $A_Z(P)$.

In the screening notation used by the endpoint calculation, let

$$x(P) = \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{72}$$

At $P = P_{\mathrm{C}}$,

$$S_{\mathrm{required}} = \begin{array}{l}\texttt{0.8954001326476587978058002831816706413077098648122916484483}\\ \texttt{0591011545242521273086888107864576}\end{array} \tag{73}$$

$$c_Q = \begin{array}{l}\texttt{0.6580257599271554356382301702323600504249200990705863960856}\\ \texttt{6007832470711257322011342309305088}\end{array} \tag{74}$$

where

$$S_{\mathrm{required}} = 1 - x + c_Q x^2. \tag{75}$$

Informally: the needed hadronic and endpoint correction can be summarized as a small second-order screening coefficient. A pure source proof has to emit this coefficient from a Ward-projected hadronic spectral measure.

Why this step is needed: hadrons are not pointlike free quarks at low energy. The coefficient is a compact way to record what the empirical hadronic spectral function contributes in this endpoint convention.

22 Step 19: Why Raw PDG $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)$ Is a Different Quantity

The direct PDG/CERN diagnostic uses

$$A_Z(P_{\mathrm{C}}) = \begin{array}{l}\texttt{128.3079654732862482099611087417567161872454761803653564600 5}\\ \texttt{34216963511778416828 5644}\end{array} \tag{76}$$

$$\Delta_{\mathrm{lep}}(P_{\mathrm{C}}) = \begin{array}{l}\texttt{4.309397866452204027131743897534489401848715660557677319471 1}\\ \texttt{5280896656803132579064661 29}\end{array} \tag{77}$$

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.02761. \tag{78}$$

It forms

$$A_L = A_Z + \Delta_{\mathrm{lep}}, \qquad A_{\mathrm{PDGdiag}} = \frac{A_L}{1 - \Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)}. \tag{79}$$

Numerically,

$$A_{\mathrm{PDGdiag}} = \begin{array}{l}\texttt{136.3828950726955771214151242189771651180022335080811544539 9}\\ \texttt{95007202025379456891237945812 89}\end{array} \tag{80}$$

This differs from 137.035999177. Holding this raw hadronic denominator shift fixed, the target would require

$$\Delta\alpha_{\mathrm{had,req}} = \texttt{0.03224434355788728882226849923695794742200734761222608150659308922187138092611795014143854 9} \tag{81}$$

or a same-scheme source-anchor bridge of

$$(\Delta\alpha_{\mathrm{had,req}} - \Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)) \cdot \alpha^{-1}(0) = \texttt{0.635071899984577762907147360708794410905808159076966220475425494682254126991352913387 1} \tag{82}$$

inverse-alpha units, with $\Delta\alpha_{\mathrm{had,req}}$ from Eq. (81), $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.02761$, and $\alpha^{-1}(0) = 137.035999177$ the CODATA central endpoint.

Informally: the raw electroweak-review hadronic running number is a useful diagnostic. The OPH endpoint needs the same electromagnetic current and the same inverse-alpha endpoint convention used by $A_Z(P)$. Mixing the two conventions moves the answer by a visible amount.

Why this step is needed: a common mistake is to insert a published $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)$ number as though it were the OPH endpoint contribution. This section shows the numerical consequence of that convention mismatch.

23 Step 20: The Measured Hadronic Spectral Input

The standard measured hadronic vacuum-polarization relation uses

$$\Delta\alpha_{\mathrm{had}}(q^2) = -\frac{\alpha q^2}{3\pi}\,\mathrm{P.V.}\int \frac{R(s)}{s(s-q^2)}\,\mathrm{d}s, \tag{83}$$

where $R(s)$ is the measured ratio of the bare $e^+e^- \to$ hadrons cross section to the pointlike $e^+e^- \to \mu^+\mu^-$ cross section.

The independently documented measured quantity in the standard electroweak literature is usually quoted as $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2)$. It is a dimensionless change in the running electromagnetic coupling at the Z-boson mass, not the inverse-alpha correction $\Delta^{\mathrm{fp}}_{\mathrm{H,req}}$ used in Eq. (66). For example, Davier, Hoecker, Malaescu, and Zhang report

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (275.7 \pm 1.0) \times 10^{-4}$$

from e^+e^--based data [12]. A perturbative update by Erler and Ferro-Hernandez gives

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (276.29 \pm 0.38 \pm 0.62) \times 10^{-4}$$

when low-energy cross-section data are used as input [15].

The comparison residual in Eq. (66) is different. It is expressed in inverse-alpha units and in the endpoint convention used by $A_Z(P)$. The standard published hadronic-running values document the physics source of the correction. They do not by themselves give the OPH same-scheme inverse-alpha correction to arbitrary precision. The community does publish the data machinery needed for this kind of calculation. Jegerlehner's alphaQED package provides hadronic running routines, covariance data, and an integration routine for custom kernels [17]. That is the right empirical route once the OPH endpoint kernel and finite same-scheme remainder are fixed.

In the OPH endpoint convention this is represented as a same-current spectral functional,

$$\Delta_{\mathrm{had}}(P) = \frac{m_Z(P)^2}{3\pi} \int \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,\mathrm{d}s, \tag{84}$$

plus the same-scheme finite remainder needed by Eq. (70).

Informally: the measured hadronic input supplies the electromagnetic spectral information required for the endpoint. The single published number 0.0276 is one weighted integral of that spectral information. The OPH endpoint needs a different weighted integral plus the same-scheme finite remainder. A pure source theorem requires the same spectral object from OPH source data.

Why this step is needed: the dispersion relation explains why measured $e^+e^- \to$ hadrons data are the right empirical input. They measure the electromagnetic spectral function that the endpoint transport requires.

24 No-Go: The 24-Slot Register Scale Does Not Determine the Hadronic Residual

Theorem 1 (Coarse 24-slot register data do not determine the Thomson residual)**.** *The specified OPH source objects P, $\alpha_U(P)$, $N_c = 3$, $N_g = 3$, and the 24-slot repair-register count do not determine the exact low-energy Thomson residual.*

Proof. The hadronic endpoint contribution depends on the Ward-projected spectral functional

$$\Delta_{\mathrm{had}}(P) = \frac{m_Z(P)^2}{3\pi} \int_0^\infty \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,\mathrm{d}s \tag{85}$$

up to the same-scheme finite endpoint remainder. Let

$$K_P(s) = \frac{m_Z(P)^2}{3\pi s[s+m_Z(P)^2]}.$$

This kernel is not constant. Two positive spectral measures can therefore have the same coarse register count, the same leading normalization, and the same electroweak branch data while giving different residuals. For example, with $s_1 \neq s_2$,

$$\rho_1(s) = M\delta(s-s_1), \qquad \rho_2(s) = M\delta(s-s_2)$$

have equal total mass, but

$$\int K_P(s)\rho_1(s)\,\mathrm{d}s = MK_P(s_1), \qquad \int K_P(s)\rho_2(s)\,\mathrm{d}s = MK_P(s_2),$$

which are generically unequal. Thus the 24-slot register scale explains the size of the correction but does not select the spectral measure that fixes its exact value. $\square$

The hadronic precision audit sharpens the no-go statement. The $\rho_Q(s;P)$ object used here is the two-current marginal of a larger source-derived hadronic spectral hadronic backend: a source QCD quotient ensemble, source parameter map, Ward-normalized current data, Stieltjes/Jacobi export, same-scheme remainder, and systematics/no-target-leak receipt bundle. That marginal can close the running-α/HVP transport only after it is emitted by the backend. It cannot be recycled as a full hadronic precision source, because HLbL and rare decay long-distance amplitudes require four-current and transition spectral data from the same source law.

Theorem 2 (Tautological 24-Jacobi realization)**.** *For any positive residual $R > 0$, there is an exact 24-point Jacobi/Stieltjes representation that reproduces R. Such a representation is non-predictive unless its nodes, weights, normalization, and finite remainder are emitted by OPH source rules before comparison with the Thomson endpoint.*

Proof. Choose any positive spectral nodes $s_1, \dots, s_{24} > 0$ and any positive normalized weights $u_j > 0$ with $\sum_j u_j = 1$. The finite inverse Stieltjes/Favard construction gives a positive 24×24 Jacobi matrix $J_{24,Q}$ whose e_1-spectral measure is

$$\mathrm{d}\mu_0(s) = \sum_{j=1}^{24} u_j \delta(s - s_j).$$

Equivalently,

$$e_1^\top f(J_{24,Q}) e_1 = \sum_{j=1}^{24} u_j f(s_j)$$

for every function f on the spectrum. Define

$$D(J_{24,Q}; m) = e_1^\top \left[J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1.$$

Since $s_j > 0$,

$$D(J_{24,Q}; m) = \sum_{j=1}^{24} u_j \left(\frac{1}{s_j} - \frac{1}{s_j + m^2} \right) > 0.$$

For any same-scheme finite remainder $\Xi_Q < R$, set

$$\omega_Q = \frac{3\pi (R - \Xi_Q)}{D(J_{24,Q}; m)}.$$

Then

$$\frac{\omega_Q}{3\pi} e_1^\top \left[J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1 + \Xi_Q = R.$$

The construction reproduces the chosen residual exactly, but the freedom in s_j, u_j, ω_Q, and Ξ_Q means it is a back-solving theorem rather than a source derivation. □

The source-side object needed for a final OPH endpoint theorem is therefore

$$R_Q(P) = \frac{\omega_Q(P)}{3\pi} e_1^\top \left[J_{24,Q}(P)^{-1} - \left(J_{24,Q}(P) + m_Z(P)^2 I \right)^{-1} \right] e_1 + \Xi_Q(P), \tag{86}$$

or equivalently the source-emitted spectral density $\rho_Q(s;P)$ plus the same-scheme finite remainder. The required dependency graph has no path from the measured Thomson endpoint into $J_{24,Q}$, ω_Q, Ξ_Q, or ρ_Q. The available execution embeds target constants, uses a mismatched P, conflates the closure residual with the total payload, and supplies a sampled envelope rather than a certified interval. It does not support a physical conclusion. A qualifying contract fixes one coordinate schema, requires a source-emitted function over the same P, certifies quadrature and interval bounds, and scores a detached artifact.

25 Step 21: The Conditional Pure Source Theorem

Theorem 3 (OPH fine-structure endpoint, conditional pure source form)**.** *Assume:*

(i) the OPH overlap-consistency branch emits the source map $P \mapsto A_Z(P)$ by Eqs. (12)–(30);

(ii) the Ward-projected $\mathrm{U}(1)_Q$ transport theorem emits a same-scheme endpoint map

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P);$$

(iii) a target-independent source law selects this source and transport map from the admitted map family;

(iv) a typed same-quantity bridge identifies $(P-\varphi)/\sqrt{\pi}$ and $1/A_{\mathrm{Th}}(P)$ as readings of one physical screen cell, with $1/A_{\mathrm{Th}}(P)$ identified as its Thomson-limit electromagnetic coupling;

(v) $G(P) = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ is a self-map and a contraction on the selected branch interval I;

(vi) the interval image contains the root and the residual bound is certified.

Then there is a unique $P_\star \in I$ satisfying $G(P_\star) = P_\star$, and the fine-structure constant on that branch is

$$\alpha(0) = \frac{1}{A_{\mathrm{Th}}(P_\star)} = \frac{P_\star - \varphi}{\sqrt{\pi}}.$$

Proof. By assumption (v), $G : I \to I$ is a contraction. Banach's fixed-point theorem gives a unique $P_\star \in I$ such that $G(P_\star) = P_\star$. Equation (11) gives

$$P_\star - \varphi = \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P_\star)}.$$

Dividing by $\sqrt{\pi}$ gives

$$\frac{P_\star - \varphi}{\sqrt{\pi}} = \frac{1}{A_{\mathrm{Th}}(P_\star)}.$$

The left side is $\alpha_{\mathrm{ext}}(P_\star)$, and the right side is $\alpha_{\mathrm{in}}(P_\star)$. Assumptions (iii)–(iv) select this map and identify their common value with the Thomson-limit electromagnetic coupling $\alpha(0)$. $\square$

Informally: a source-derived hadronic spectral endpoint map would supply the open transport step. A physical fine-structure conclusion also requires target-independent map selection and the same-quantity bridge. With those premises and the interval proof, the selected full source map has one fixed point and the bridge identifies its value with the fine-structure constant.

The interval form of assumptions (v)–(vi) is direct: on any interval I where interval-arithmetic evaluation certifies $G(I) \subseteq I$ together with a derivative bound $|G'| \leq L < 1$, the Banach fixed-point theorem gives both existence and uniqueness of the fixed point in I. The public interval contraction certificate [4] proves $G(I) \subseteq \mathrm{int}(I)$ and $L \leq 0.0724$ for both readout modes by a direct Banach argument in mean-value (centered) form, with `mpmath.iv` outward rounding on every elementary operation and with the SU(2)/SU(3) edge-sum tails bounded by geometric majorants, so the enclosure covers the infinite-cutoff sums. The certified unique fixed points are $\alpha^{-1} = 136.994835177413\ldots$ for the source map (enclosure width 7.2×10^{-24}) and $\alpha^{-1} = 137.035660136946577\ldots$ for the gauge-width map. The certificate certifies the declared numerical map; it does not relate either fixed point to the measured constant. Both roots lie outside the measured comparison basin. The global at-most-one statement is discharged on the declared analytic domain ($\alpha^{-1} \in [100, 200]$): $\sup |g'| < 1$ on all 256 certified pieces, both readout maps, zero exceptional set (domain-global uniqueness certificate in the same artifact bundle), so each map has exactly one fixed point on the domain. A maximal-domain extension certificate closes the exterior. An envelope lemma built from the declared solver windows (the pixel scan window $\alpha_U \in [0.02, 0.08]$ and the m_Z log-grid window $\ln(\mu_U/m_Z) \in [0, 50]$) together with the m_Z-closure identity certifies $1/\alpha_3(m_Z) \geq 4$ and a positive inverse-alpha readout floor on the whole analytic domain, and an interval sweep of the full pixel window certifies that every window-consistent fixed-point candidate of either readout map lies inside the declared solver interval (global uniqueness extension certificate). Each declared map therefore has exactly one fixed point on its maximal analytic domain, and the set of fixed points outside the declared interval is empty; the headline enclosures above are unchanged.

A source-only theorem requires a source-derived Ward-current spectral measure, the source Jacobi kernel, electromagnetic normalization, a same-scheme finite remainder, a target-independent rule selecting one physical map, a typed construction that identifies its two sides as the same quantity, a dependency graph excluding Thomson-target leakage, a full contraction interval for the selected map, and a prediction interval narrower than the adopted direct measurement. For the comparison standard used here, the direct rubidium recoil result of Morel, Yao, Cladé, and Guellati-Khélifa reports $\alpha^{-1} = 137.035999206(11)$ with 81 parts-per-trillion relative precision [14]. The operational receipt is therefore

$$\mathrm{rad}(\alpha^{-1}_{\mathrm{pred}}) < 1.1 \times 10^{-8}$$

in inverse-alpha units for the cited direct rubidium recoil interval.

26 Step 22: The CODATA/NIST Comparison Endpoint

With the CODATA/NIST measured endpoint value

$$A_{\mathrm{Th}}(P_{\mathrm{C}}) = A_{\mathrm{C}} = 137.035999177(21), \tag{87}$$

Eq. (11) gives the corresponding comparison pixel:

$$\boxed{P_{\mathrm{C}} = \begin{array}{l}\texttt{1.630968209403959324879279847782648941335982851627925060666}1\\ \texttt{507533907793398933432}\end{array}} \tag{88}$$

$$\boxed{\alpha(0) = \begin{array}{l}\texttt{0.0072973525643314250302457952646916832280660213133653604957}\\ \texttt{798803819933561573639928}\end{array}} \tag{89}$$

$$\boxed{\alpha^{-1}(0) = 137.035999177(21).} \tag{90}$$

Informally: the endpoint number is the same number NIST/CODATA reports. The display exposes the OPH-side bookkeeping around that measured value. A source theorem would additionally need a target-independent map choice, the same-quantity bridge, and the low-energy same-scheme hadronic calculation.

27 Reproducibility

The executable derivation lives in the public repository under https://github.com/FloatingPragma/observer-patch-holography/tree/main/code/P_derivation. The main human-facing CLI is

```
cd reverse-engineering-reality/code/P_derivation
python3 derive_p.py --color always
```

To print only the pure source value:

```
python3 derive_p.py --no-hadron-closure
```

To emit machine-readable output:

```
python3 derive_p.py --json --output runtime/report.json
```

To run the raw PDG/CERN diagnostic discussed in Section 19:

```
python3 fine_structure_fixed_point_demo.py --compare-alpha-inv 137.035999177
```

Informally: the command line separates the pure source value, the CODATA/NIST comparison value, and the diagnostic values. The mathematical paper uses the same separation.

28 Complete Source-to-Fixed-Point Chain

Step	Mathematical object	Scope
Overlap rule	Eq. (1)	OPH starting structure.
Pixel variable	$P = a_{\mathrm{cell}}/\ell_\star^2$	Defined.
Golden balance	$\varphi = (1+\sqrt{5})/2$	Defined.
Boundary width	$\alpha_{\mathrm{ext}} = (P-\varphi)/\sqrt{\pi}$	Defined.
Inside readout	$\alpha_{\mathrm{in}} = 1/A_{\mathrm{Th}}(P)$	Defined.
Closure	$P = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$	Defined.
Source scale	$M_U = E_P e^{-2\pi P^{1/6}}$	Defined and calculated.
Cell scale	$E_{\mathrm{cell}} = E_P/\sqrt{P}$	Defined and calculated.
Transmutation	$v = E_{\mathrm{cell}} \exp[-2\pi/(\beta_{\mathrm{EW}}\alpha_U)]$	Defined and calculated.

Step	Mathematical object	Scope
Gauge running	Eq. (17)	Defined and calculated; the coefficient triple is the one-loop MSSM set, a counted structural selection.
Z-scale self-consistency	Eq. (20)	Defined and calculated.
Heat-kernel closure	Eq. (27)	Defined and calculated.
Electroweak anchor	$A_Z = \alpha_{\rm em}^{-1}(m_Z^2; P)$	Defined and calculated.
Charged spectrum	Eqs. (36) and (39)	Declared phenomenological transport input; not an OPH-derived mass prediction.
Fermion kernel	Eqs. (40) and (42)	Exact one-loop kernel.
Screening ansatz	Eq. (46)	Declared selection of the quark continuation; asserted rather than derived, with no perturbative derivation.
Calculated endpoint	$A_{\rm calc} = A_Z + \Delta_{\rm calc}$	Raw no-hadron diagnostic conditional on the supplied charged-spectrum continuation.
Bare-α_U closure	$A^{\rm fp}_{\alpha_U}$, Eq. (63)	Mixed-provenance display packet; the certified gauge-width fixed point is $137.035660136946577\ldots$, without constituting a derivation of α.
Required hadronic correction	$\Delta^{\rm fp}_{\rm H,req}$, Eq. (66)	CODATA/NIST comparison residual attributed to the missing same-scheme hadronic endpoint transport.
Hadronic/scheme residual	$R_Q = A_{\rm Th} - A_{\rm calc}$	Isolated endpoint contribution.
CODATA/NIST endpoint	$A_{\rm Th} = 137.035999177(21)$	Measured endpoint comparison value.

Step	Mathematical object	Scope
Two-P provenance	P_{root} versus P_{C}	Source-branch root and measured-endpoint comparison pixel are distinct; P_{C} inherits CODATA/NIST by definition.
Raw PDG diagnostic	Eq. (79)	Comparison diagnostic in a different endpoint convention.
Interval theorem	Banach/contraction certificate for full G	Discharged for the declared numerical map (public certificate, both readout modes); required for the full source map with the hadronic term.

29 Conclusion

Conditional on target-independent map selection, the declared same-cell bridge, and same-scheme endpoint transport, the fine-structure constant is the coupling that makes one OPH screen cell internally consistent. The outside reading says that the cell is displaced from the golden-ratio balance by $(P-\varphi)/\sqrt{\pi}$. The inside reading sends the same P through the declared source map, gauge closure, electroweak anchor, and Ward-projected electromagnetic transport, producing $1/A_{\mathrm{Th}}(P)$. Neither the exact fixed-point mathematics nor the source certificates construct that bridge or select either certified map. The construction carries zero continuous dials and each declared incomplete map has a machine-certified unique fixed point on its declared domain. The map selector, same-cell construction, hadronic source payload, and detached no-target-leak certificate are premises this construction does not supply, so the residuals are diagnostics rather than a derivation of α.

The pure source calculation computes the root witness

$$\alpha_{\mathrm{root}}^{-1} = \begin{array}{l}\texttt{136.99483517741293729528942946443690285756120615103539318450}\\ \texttt{2}\end{array},$$

the interval-certified unique fixed point of the declared source map (enclosure width 7.2×10^{-24}, shipped interval certificate). The CODATA/NIST measured endpoint value is

$$\alpha^{-1}(0) = 137.035999177(21),$$

with

$$P = 1.63096820940395932487927984778264894 1\ldots.$$

Combining the source root with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel gives the mixed-provenance no-hadron diagnostic

$$A_{\alpha_U}^{\mathrm{fp}} = 137.03595951360856779028942946443 7\ldots.$$

This mixed-provenance diagnostic combines the certified source root with α_U at the CODATA-derived comparison pixel and is not a fixed point of a single declared map. The certified self-consistent gauge-width fixed point is $\alpha^{-1} = 137.035660136946577\ldots$, 2.5×10^{-6} below the measured value, about 1.6×10^4 measurement sigma. The residual gap is CODATA-minus-diagnostic bookkeeping, not an emitted low-energy hadronic calculation. The physical pure-source theorem requires the Ward-projected hadronic spectral measure, same-scheme transport, target-independent map selection, the typed same-quantity bridge, and the interval-certificate step.

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Code Availability declaration. The fixed-point code and runtime artifacts are available at https://github.com/FloatingPragma/observer-patch-holography/tree/main/code/P_derivation.

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References

[1] B. Müller, A. Osika, M. Poneder, K. Xue, B. Cassie, P. Nguyen, M. A. Visser, K. A. Anirudha, D. Matscheko, and J. Hill, *From Observer Consensus to Standard Physics*. https://wkaxfdgxoqmghwgshymt.supabase.co/storage/v1/object/public/papers/from_observer_consensus_to_standard_physics.pdf.

[2] B. Müller, A. Osika, M. Poneder, K. Xue, P. Nguyen, M. A. Visser, and D. Matscheko, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*. GitHub source: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.tex. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf.

[3] B. Müller, A. Osika, M. Poneder, K. Xue, M. A. Visser, and D. Matscheko, *Deriving the Particle Zoo from Observer Consistency*. GitHub source: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.tex. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[4] OPH executable derivation code, https://github.com/FloatingPragma/observer-patch-holography/tree/main/code/P_derivation. Primary files: paper_math.py, derive_p.py, fine_structure_fixed_point_demo.py.

[5] OPH hadron data policy, https://github.com/FloatingPragma/observer-patch-holography/blob/main/docs/HADRON.md.

[6] U. D. Jentschura and I. Nandori, *Attempts at a determination of the fine-structure constant from first principles: A brief historical overview*, Eur. Phys. J. H **39**, 591–613 (2014). https://arxiv.org/abs/1411.4673.

[7] R. Heyrovska and S. Narayan, *Fine-structure Constant, Anomalous Magnetic Moment, Relativity Factor and the Golden Ratio that Divides the Bohr Radius*. https://arxiv.org/abs/physics/0509207.

[8] T. P. Singh, *Quantum theory without classical time: octonions, and a theoretical derivation of the fine structure constant* 1/137, Int. J. Mod. Phys. D **30**, 2142010 (2021). https://arxiv.org/abs/2110.07548.

[9] J. Ciborowski, *Bi-Constructible pattern of weak and flavour mixing: implications for electroweak coupling constants*. https://arxiv.org/abs/2508.00030.

[10] S. V. Chekanov and H. Kjellerstrand, *Discovering the Underlying Analytic Structure Within Standard Model Constants Using Artificial Intelligence*, Particles **8**, 95 (2025). https://arxiv.org/abs/2507.00225.

[11] Q. Liu, I. Low, and Z. Yin, *A Quantum Computational Determination of the Weak Mixing Angle in the Standard Model*. https://arxiv.org/abs/2509.18251.

[12] M. Davier, A. Hoecker, B. Malaescu, and Z. Zhang, *Reevaluation of the Hadronic Contributions to the Muon $g-2$ and to $\alpha(M_Z)$*, Eur. Phys. J. C **71**, 1515 (2011). https://arxiv.org/abs/1010.4180.

[13] A. Keshavarzi, D. Nomura, and T. Teubner, *The $g-2$ of charged leptons, $\alpha(M_Z^2)$ and the hyperfine splitting of muonium*, Phys. Rev. D **101**, 014029 (2020). https://arxiv.org/abs/1911.00367.

[14] L. Morel, Z. Yao, P. Cladé, and S. Guellati-Khélifa, *Determination of the fine-structure constant with an accuracy of 81 parts per trillion*, Nature **588**, 61–65 (2020). https://doi.org/10.1038/s41586-020-2964-7.

[15] J. Erler and R. Ferro-Hernandez, *Perturbative contributions to $\Delta\alpha^{(5)}(M_Z^2)$*. https://arxiv.org/abs/2308.05740.

[16] Particle Data Group, *Data files and plots of cross-sections and related quantities in the Review of Particle Physics*. https://pdg.lbl.gov/current/xsect/.

[17] F. Jegerlehner, *Software packages alphaQED and pQCDAdler*. https://people.physik.hu-berlin.de/~fjeger/software.html.

[18] B. Müller, *Fine-Structure Constant from First Principles*, Floating Pragma Blog. https://blog.floatingpragma.io/fine-structure-constant-from-first-principles.

[19] NIST, *CODATA Value: inverse fine-structure constant*, 2022 CODATA recommended value. https://physics.nist.gov/cgi-bin/cuu/Value?eqalphinv=.

[20] P. J. Mohr, E. Tiesinga, D. B. Newell, and B. N. Taylor, *CODATA Recommended Values of the Fundamental Physical Constants: 2022*. NIST publication page: https://www.nist.gov/publications/codata-recommended-values-fundamental-physical-constants-2022.

[21] NIST/CODATA, *CODATA Recommended Values of the Fundamental Physical Constants: 2022*, wall chart. https://physics.nist.gov/cuu/pdf/wall_2022.pdf.